Reproducing and Stress-Testing PerSAM for One-Shot Personalized Segmentation

Ian Yang (qy225) Yawen Lin (yl3646)

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1 Introduction

The Segment Anything Model (SAM) is a powerful promptable segmentation model, but it still requires a user prompt for each target image. This is limiting when a user wants to repeatedly segment the same personalized object, such as a specific toy, pet, or household item, across many images. Zhang et al. address this problem in *Personalize Segment Anything Model with One Shot* by proposing PerSAM, which personalizes SAM from only one reference image and one reference mask [1]. Given this one-shot input, PerSAM extracts a target representation, localizes similar regions in new images, and guides SAM to segment the user-specified object. The paper also proposes PerSAM-F, a lightweight variant that freezes SAM and tunes only two mask-scale weights to reduce part-vs-whole mask ambiguity.

Our project reproduces the core PerSeg personalized segmentation result and adds an independent robustness experiment. We ask whether PerSAM still works when the reference mask is imperfect, since real users may provide rough, noisy, or spatially misaligned masks.

2 Chosen Result

We chose to reproduce the paper's main PerSeg personalized object segmentation result, which is the most direct test of the paper's central contribution: whether SAM can be personalized to segment a specific visual concept from a single reference image-mask pair. We focus on PerSAM and PerSAM-F rather than the video or DreamBooth extensions because the PerSeg evaluation provides a clean quantitative benchmark for one-shot personalized segmentation.

The original paper reports that PerSAM reaches about 89.3 mIoU on PerSeg, while PerSAM-F improves to about 95.3 mIoU through lightweight scale-aware fine-tuning [1]. We aimed to reproduce this PerSeg quantitative comparison using the official implementation, the PerSeg dataset, the SAM ViT-H checkpoint, and the provided mIoU evaluation script.

3 Methodology

We used the official PerSAM codebase as the basis for our re-implementation. Our pipeline was: download and organize the PerSeg dataset into `Images/` and `Annotations/`; run PerSAM and PerSAM-F in Google Colab using an L4 GPU and the SAM ViT-H checkpoint; and evaluate predicted masks using `eval_miou.py`. The evaluation metrics are mean Intersection-over-Union (mIoU) and mean accuracy (mAcc).

PerSAM is training-free. For each object category, it uses the reference image and mask indexed by `00`, computes a similarity-based location prior in the target image, selects positive and negative prompt locations, and guides SAM using target information. PerSAM-F starts from the same pipeline but freezes SAM and tunes a small number of mask-scale weights, which helps choose the correct object scale when SAM produces multiple plausible masks.

For our independent experiment, we modified only the reference mask while keeping all evaluation ground-truth masks unchanged. We tested four perturbations: erosion, dilation, spatial shift, and random pixel noise. This isolates how imperfect one-shot supervision affects PerSAM. A key implementation issue was mask encoding: some PerSeg foreground masks use value 38 rather than 255. A naive threshold at 127 erased the object, so our perturbation script binarized masks using `mask > 0`.

4 Results & Analysis

Table 1 shows that our reproduction matches the main trend of the paper. PerSAM achieves 89.32 mIoU and 92.19 mAcc. PerSAM-F improves to 95.18 mIoU and 95.57 mAcc, a +5.86 mIoU gain.

Table 1: Main reproduction results on PerSeg.

Method	mIoU	mAcc
PerSAM	89.32	92.19
PerSAM-F	95.18	95.57
Improvement	+5.86	+3.38

The improvement is driven by hard PerSAM failure cases. For example, `can` improves from 38.91 to 97.49 IoU, `rc_car` from 39.30 to 96.12, `teapot` from 40.02 to 96.93, and `robot_toy` from 60.55 to 96.66. This supports the paper's claim that PerSAM-F helps when PerSAM chooses an incorrect mask scale or confuses a local part with the whole object. However, PerSAM-F is not uniformly better. For `colorful_teapot`, IoU drops from 96.27 to 84.48. This object is already visually distinctive for PerSAM; the added scale tuning may over-correct around thin structures such as the handle and spout.

Table 2 summarizes our robustness experiment. Erosion and dilation cause only small drops, and sparse random noise has no measurable effect. In contrast, shifting the reference mask sharply decreases mIoU from 89.32 to 73.84. This suggests that PerSAM is robust to imperfect mask boundaries but sensitive to spatial misalignment. Since the target embedding is extracted from the masked reference region, a shifted mask can contaminate the representation with background or wrong-object features.

Table 2: PerSAM robustness to imperfect reference masks.

Reference Mask	mIoU	mAcc	Δ mIoU
Original	89.32	92.19	–
Erode	88.45	92.18	-0.87
Dilate	87.33	90.23	-1.99
Shift	73.84	82.28	-15.48
Noise	89.32	92.19	0.00

5 Reflections

This project showed that reproducing a foundation-model paper is not only about running the main script. Much of the work involved setting up data paths, checkpoints, evaluation scripts, and mask formats correctly. The mask-value bug was especially important: using `mask > 127` destroyed masks whose foreground value was 38, while `mask > 0` fixed the perturbation pipeline.

The main takeaway is that PerSAM efficiently turns SAM into a one-shot personalized segmenter. PerSAM-F improves many hard cases by resolving scale ambiguity, but it can hurt easy objects where PerSAM already works well. Our robustness experiment adds a practical lesson: boundary quality matters less than reference-mask alignment. Future work could test multiple reference examples, automatic reference-mask correction, or stronger real-world user annotation noise.

6 References

References

[1] Renrui Zhang, Zhengkai Jiang, Ziyu Guo, Shilin Yan, Junting Pan, Xianzheng Ma, Hao Dong, Peng Gao, and Hongsheng Li. *Personalize Segment Anything Model with One Shot*. arXiv:2305.03048, 2023.

[2] Alexander Kirillov et al. *Segment Anything*. ICCV, 2023.

[3] PerSAM official repository. `https://github.com/ZrrSkywalker/Personalize-SAM`.

[4] Segment Anything official repository. `https://github.com/facebookresearch/segment-anything`.